

Lab 4: Developer Productivity Agent

Alignment: Domains 2 and 5 - Native Tools and Context Management Estimated time: 75 minutes Mode: Local simulation; no API key required

Objective

Explore a codebase with a narrow Glob-Grep-Read sequence, record findings in an external scratchpad, and avoid whole-repository ingestion.

Scenario

A developer asks where invoice totals are calculated and validated. The agent must locate the relevant path and dependency with a small, reproducible evidence set.

Prerequisites

- Python 3.10 or newer.
- A terminal opened in this lab folder.
- No production account, network access, or secret is required.

Architecture before execution

Read the source and identify the control boundary, the evidence artifact, and the terminal condition before running it. All side effects are confined to this lab's evidence/ folder.

Procedure

1. Inspect the sample codebase without opening every file. `bashfind sample_repo -maxdepth 3 -type f | sort`### 2. Run the exploration workflow for calculate_total.` `bashpython3 -B explorer.py calculate_total`### 3. Inspect the retrieval trace and note the progressive expansion from path match to symbol match to focused read.` `bashpython3 -m json.tool evidence/retrieval.json`### 4. Review the scratchpad. Confirm it records decisions and evidence without copying the entire source tree.` `bashsed -n '1,200p' evidence/scratchpad.md`### 5. Run the verifier and save its output.` `bashmkdir -p evidence && python3 -B verify.py | tee evidence/verification.txt``

Acceptance checks

- [] Only text-like candidate files are searched.
- [] Focused reads include matched lines and a small context window.
- [] The dependency on `validate_items` is captured.
- [] The scratchpad records source paths and open questions.

Reflection

What evidence would justify expanding the read set, and what signal would tell you to stop retrieving more context?

Evidence and completion

Save terminal output in evidence/verification.txt. The lab is complete when the verifier prints PASS, every acceptance check is satisfied, and your reflection identifies one design trade-off.

Troubleshooting

- Run commands from this lab folder so relative paths resolve correctly.
- Use python3 -B to prevent learner machines from creating __pycache__ artifacts.
- If a check fails, read the named failed assertion, inspect the referenced file, and rerun only after correcting the underlying design.

Clean-up

Remove only files you created outside this folder. Never store API keys, access tokens, customer data, or production logs in lab evidence.